

Notes: Rajan and Zingales (JF, 1995)

What Do We Know about Capital Structure? Some Evidence from International Data

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1 Big picture

- **Research question.** Are the capital-structure regularities documented for U.S. firms also present in other major industrialized countries? More broadly, what do we really know about how firms choose leverage?
- **Core motivation.** Theories of capital structure emphasize taxes, bankruptcy costs, asymmetric information, agency problems, and institutional environments. But much of the empirical evidence before this paper comes from U.S. data. Rajan and Zingales use international evidence to test whether the usual U.S. patterns are robust.
- **Main empirical setting.** The paper studies nonfinancial public firms in the G-7 countries: the United States, Japan, Germany, France, Italy, the United Kingdom, and Canada.
- **Main data source.** The firm-level data come from Global Vantage, with the main analysis focused on 1987–1991 to reduce backfill and survivorship concerns.
- **Central measurement issue.** International leverage comparisons are hard because accounting rules differ across countries. The paper therefore spends substantial effort making leverage measures more comparable.
- **Main aggregate finding.** Once leverage is measured carefully, firms in G-7 countries look more similar than earlier work suggested. The United Kingdom and Germany are relatively less levered, but there is no simple sharp division between “bank-oriented” and “market-oriented” systems in leverage levels.
- **Main firm-level finding.** The same variables that are correlated with leverage in U.S. data are generally correlated with leverage in other countries: tangibility is positively related to leverage, market-to-book is negatively related, size is usually positively related, and profitability is usually negatively related.
- **Interpretive twist.** The cross-country robustness of these correlations is striking, but the paper argues that the theoretical interpretation remains unresolved. Similar coefficients do not necessarily mean that we understand the economic mechanism.
- **Economic takeaway.** Capital-structure facts appear robust internationally, but simple stories based only on taxes, bankruptcy law, or bank-versus-market financial systems are not enough to explain observed leverage.

2 Incremental contribution of the paper

- **International test of U.S. capital-structure facts.** The paper takes well-known U.S. determinants of leverage and asks whether they survive in other institutional environments.
- **Careful accounting adjustments.** A major contribution is measurement. The authors show that naive international leverage comparisons can be misleading because countries differ in consolidation, pension accounting, leases, provisions, taxes, and asset valuation conventions.
- **Reassessment of aggregate leverage differences.** Earlier studies often suggested large leverage differences across countries. Rajan and Zingales show that, after adjustments, aggregate leverage is more similar across G-7 countries than previously thought.

- **Institutional interpretation.** The paper does not merely run regressions. It discusses taxes, bankruptcy law, financial-market structure, and financing flows to ask whether institutional differences can explain capital-structure patterns.
- **Warning against over-interpretation.** The paper emphasizes that the common empirical factors are not yet tightly mapped to theory. For example, size is often interpreted as an inverse proxy for bankruptcy costs, but the German evidence does not fit that interpretation cleanly.
- **Foundation for later capital-structure research.** The paper becomes a benchmark for international corporate finance by showing both what is robust and what remains theoretically ambiguous.

3 Data and measurement

3.1 Global Vantage firm sample

The paper uses Global Vantage, a database of international public corporations. The main sample consists of nonfinancial firms in the G-7 countries. Financial firms are excluded because their leverage is strongly shaped by deposit insurance, regulation, and financial-intermediary balance-sheet structure.

- **Countries.** United States, Japan, Germany, France, Italy, United Kingdom, and Canada.
- **Main period.** 1987–1991. The authors avoid relying heavily on earlier backfilled data.
- **Firm type.** Publicly traded nonfinancial firms.
- **Consolidation.** The main balance-sheet comparisons focus on firms reporting consolidated balance sheets, because unconsolidated accounts can understate leverage when debt is held in affiliates.
- **Sample coverage.** Global Vantage covers a large fraction of listed-firm market capitalization in each country, but the sample is tilted toward larger listed firms and may not represent all firms in the economy.

Interpretation. The sample is useful for comparing large public firms across developed economies. It is less suited for statements about small private firms, which may rely more heavily on banks and private debt.

3.2 Leverage measures

The paper emphasizes that leverage has no single universal measure. Different theories call for different concepts of debt and assets.

- **Nonequity liabilities to total assets.** A broad measure of all obligations relative to assets.
- **Debt to total assets.** Short-term plus long-term debt divided by total assets.
- **Debt to net assets.** Debt divided by assets net of accounts payable and other current liabilities.

- **Debt to capital.** Debt divided by debt plus equity.
- **Book versus market values.** Equity can be measured at book value or market value, giving different leverage ratios.
- **Adjusted leverage.** The authors adjust for items such as cash, pension liabilities, intangibles, deferred taxes, and provisions to improve cross-country comparability.

A representative firm-level specification in the paper can be summarized as:

$$\text{Leverage}_i = \alpha + \beta_1 \text{Tangibility}_i + \beta_2 \text{Market-to-book}_i + \beta_3 \log(\text{Sales}_i) + \beta_4 \text{Profitability}_i + \varepsilon_i.$$

3.3 Firm-level explanatory variables

The firm-level regressions focus on variables that prior U.S. evidence identifies as important:

- **Tangibility.** Fixed assets divided by total assets. Tangible assets can serve as collateral, so tangibility is expected to be positively related to leverage.
- **Market-to-book.** A proxy for growth opportunities and valuation. Firms with high growth opportunities may avoid debt because distress or debt overhang is more costly.
- **Size.** Log sales. Larger firms may be more diversified and face lower expected distress costs.
- **Profitability.** EBITDA divided by book assets. Pecking-order logic predicts profitable firms may use less debt because internal funds are available.

3.4 Institutional variables and country comparisons

The paper also examines country-level institutions and financing environments:

- tax treatment of interest, dividends, retained earnings, and capital gains;
- bankruptcy procedures and creditor rights;
- bank credit, stock market capitalization, and bond market capitalization;
- aggregate sources of external financing.

Interpretation. The paper uses institutions to evaluate candidate theories rather than to estimate a fully causal model of leverage choice.

4 Methodology and empirical specifications

4.1 Making international leverage comparable

The first empirical step is measurement. The authors compare balance sheets across countries, identify major accounting differences, and construct multiple leverage ratios. This matters because international balance sheets differ in:

- consolidation of subsidiaries;

- treatment of pension liabilities;
- lease capitalization;
- provisions and reserves;
- asset revaluation rules;
- reporting of deferred taxes and intangibles.

The main idea is that an observed difference in book leverage may reflect accounting rules rather than financing policy. The paper therefore reports many leverage definitions rather than relying on a single ratio.

4.2 Aggregate leverage and financing flows

The paper compares leverage at several levels:

- median firm leverage within each country;
- aggregate leverage obtained by summing across firms;
- book versus market leverage;
- adjusted leverage after removing or reclassifying accounting items;
- external financing flows over 1984–1991.

This distinction is important because a country may have similar leverage levels but different financing flows. For example, firms can maintain similar leverage while relying more or less on new debt issuance, equity issuance, or retained earnings.

4.3 Institutional comparison

The paper reviews three main institutional channels:

- **Taxes.** Debt may be tax advantaged, but the magnitude depends on both corporate and personal taxes and on the identity of the marginal investor.
- **Bankruptcy law.** Creditor rights, automatic stays, and management control differ sharply across countries.
- **Bank versus market orientation.** Countries differ in the importance of bank loans, stock markets, and bond markets.

The paper’s conclusion is deliberately cautious: these institutional differences are real, but they do not generate a simple one-to-one explanation for leverage levels.

4.4 Firm-level leverage regressions

The final empirical step estimates within-country regressions of leverage on firm characteristics. The dependent variable is book or market leverage in 1991, while explanatory variables are four-

year averages over 1987–1990. The regressions are estimated as censored Tobit models because leverage is bounded.

The question is whether the same factors that appear important in the United States have similar signs and explanatory power abroad.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies robust cross-sectional correlations between firm characteristics and leverage across countries. It also documents broad similarities and differences in aggregate leverage, financing flows, taxes, bankruptcy regimes, and capital markets.

It does *not* identify a clean causal effect of taxes, bankruptcy law, or bank orientation on leverage. The evidence is best read as disciplined descriptive evidence and theory evaluation.

5.2 Why accounting comparability is central

A major identification problem is that leverage ratios are partly accounting artifacts. For example, a country that reports pension liabilities or provisions differently may appear more levered even if its firms have similar economic obligations. The paper’s contribution is to show that careful adjustments can materially change the cross-country ranking.

5.3 Sample selection and external validity

Global Vantage covers listed firms, and listed firms are not equally representative across countries. Public companies may be a larger or smaller share of the corporate sector depending on the country. Therefore, the paper’s results are most directly about large public firms, not all firms.

5.4 Interpreting taxes

The paper stresses that tax incentives depend on the marginal investor. A tax-exempt investor, a high-tax individual investor, and an average household can face very different relative tax advantages of debt. This makes it hard to draw simple conclusions from statutory corporate tax rates alone.

5.5 Interpreting bankruptcy law

Bankruptcy law affects capital structure through liquidation costs, creditor rights, renegotiation, and management control. But the observed country patterns are not always what simple theories predict. Germany, for instance, is often described as having creditor-friendly bankruptcy rules, yet German firms appear relatively less levered in several measures.

5.6 Theoretical ambiguity

The paper's most important interpretive message is that the empirical signs are robust but the theory behind them remains unsettled. A variable such as size, tangibility, or profitability can proxy for several mechanisms at once, and different institutional environments do not isolate the mechanism cleanly.

6 Tables and figures

6.1 Table I: Distribution of firms by size

Read:

- Table I sorts consolidated G-7 firms into size deciles using the 1991 market value of assets.
- The U.S., U.K., and Canadian samples contain more firms below the overall median than Japan, Germany, France, and Italy.
- Japan is the major outlier: almost all Japanese firms in the sample are above the overall median.
- The table warns that the sample is useful for large public firms but is not a random sample of all firms in each country.

Economic message: International comparisons of capital structure must account for differences in the size and public-listing composition of the sample.

6.2 Table II: Balance sheets across countries

Read:

- Table II compares average balance sheets for consolidated nonfinancial firms in 1991.
- Total liabilities relative to book assets differ, but the differences are less dramatic than a naive view of bank-based versus market-based systems might imply.
- Current liabilities, accounts payable, deferred taxes, pension liabilities, and other liabilities vary across countries, which affects leverage measurement.
- The table motivates the adjusted leverage measures in Table III.

Economic message: Before asking why leverage differs across countries, the paper first asks whether the accounting numbers are comparable.

Table I
Distribution of Firms Followed by Global Vantage by Size
All consolidated firms in all G-7 countries are pooled and they are placed in size deciles according to the 1991 market value of their assets in U.S. dollars.

Decile	Country (Percentage of Consolidated Firms ^a)						
	United States	Japan	Germany	France	Italy	United Kingdom	Canada
Smallest	15	0	1	9	3	4	8
2	12	0	5	2	7	11	13
3	11	0	5	8	8	17	10
4	11	0	13	4	14	13	13
5	10	3	11	11	14	12	12
6	10	5	14	13	11	12	12
7	8	16	13	17	15	10	11
8	8	22	13	13	14	8	9
9	8	25	13	13	10	7	9
Largest	8	30	11	9	4	7	3
Total number of firms	2583	514	191	225	118	608	318

^a Columns may not sum to 100 because of rounding errors.

Table I: Distribution of firms followed by Global Vantage by size.

Table II
Balance Sheets for Non-Financial Firms in the G7 Countries—1991
The value of each item is calculated as a fraction of the book value of total assets and then averaged across all firms reporting consolidated balance sheets in the country. Only balance sheets of non-financial firms are included.

	United States	Japan	Germany	France	Italy	United Kingdom	Canada
ASSETS							
Cash and short-term investments	11.2	18.4	8.8	10.3	10.5	11.4	8.2
Account receivable/debtors	17.8	22.5	26.9	28.9	29.0	22.1	13.0
Inventories	16.1	13.9	23.6	17.4	15.6	17.7	11.0
Current assets—other	2.9	3.0	0.1	1.7	1.6	3.7	1.9
Current assets—total	48.0	57.7	59.4	58.3	56.5	54.7	33.2
Fixed assets (tangible)	36.3	28.7	32.7	24.4	32.4	41.3	51.6
Investments and advances—equity	1.4	1.4	1.4	3.4	1.9	1.5	4.8
Investment and advances—other	3.1	9.4	3.4	4.9	4.1	1.2	2.9
Intangible assets	7.6	0.8	2.4	8.5	2.6	0.9	4.7
Assets—other	5.8	2.9	0.7	0.7	3.3	0.5	3.7
Assets—total	100.0	100.0	100.0	100.0	100.0	100.0	100.0
LIABILITIES							
Debt in current liabilities	7.4	16.4	9.9	11.6	16.2	9.6	7.3
Accounts payable/creditors	15.0	15.4	11.5	17.0	14.7	13.7	13.3
Current liabilities—other	11.0	10.4	8.7	17.0	12.2	16.7	2.8
Current liabilities—total	33.4	42.2	30.0	43.4	43.2	40.0	23.1
Deferred taxes	3.2	0.1	0.8	1.3	1.5	0.9	4.4
Long-term debt	23.3	18.9	9.8	15.7	12.1	12.4	28.1
Minority interest	0.6	0.9	1.6	3.9	3.4	1.1	2.0
Reserves—untaxed	0.0	0.0	1.7	0.0	0.0	0.0	0.0
Liabilities—other	5.8	4.8	28.7	6.3	7.8	3.4	2.6
Liabilities—total	66.1	66.8	72.0	68.8	67.4	57.8	60.3
Shareholders equity	34.1	33.2	28.0	31.2	32.6	42.2	39.7
Total liabilities and shareholders equity	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Source: Global Vantage Data Base.

Table II: Balance sheets for nonfinancial firms in the G-7 countries.

6.3 Table III: Leverage and adjusted leverage

Read Panel A:

- Panel A reports several leverage ratios: nonequity liabilities to assets, debt to total assets, debt to net assets, debt to capital, and interest coverage.
- It reports both book and market versions where appropriate.
- The results show that G-7 leverage levels are more similar than prior broad claims often implied.

Read Panel B:

- Panel B adjusts leverage for cash, pension liabilities, intangibles, deferred taxes, and other accounting items.
- The adjustments reduce the scope for cross-country accounting differences to drive the results.
- Germany and the United Kingdom still look relatively less levered on several measures.

Read Panel C:

- Panel C checks alternative samples and compares the Global Vantage leverage measures to OECD-based studies.
- It also examines changes in leverage over time for firms continuously reporting through 1982–1991.
- These robustness checks help explain why earlier studies reached different conclusions.

Economic message: There is no single correct leverage ratio. The core conclusion is robust only after studying a family of ratios and adjustments.

Table III
Extent of Leverage and Adjusted Leverage in Different Countries
Leverage measures are calculated for all non-financial companies reporting consolidated balance sheets in 1991. In the 'book' column, equity is measured as book value. In the 'market' column, equity is measured at market value. For instance, this alters the value of assets as follows: the (quasi) market value of assets is obtained by subtracting the book value of common equity from book assets and adding back the market value of common equity. In Panel A Non-equity liabilities to total assets is the sum of all liabilities divided by the value of assets. Debt to total assets is the value of short term plus long term debt divided by total assets. Debt to net assets is the book value of debt divided by net assets where net assets is assets minus accounts payable and other current liabilities. Debt to capital is the book value of debt divided by the sum of the book value of debt and equity. EBIT is earnings before interest and taxes. EBITDA is earnings before interest, taxes, and depreciation. Aggregate ratios are obtained by summing the numerator across all reporting firms in the country and dividing by the denominator summed across the same firms. In Panel B, Adjusted liabilities are total liabilities less pension liabilities (in Germany) less cash. In Panels B and C, Adjusted debt is the book value of debt less the value of cash and marketable securities. Adjusted assets are total assets less cash and short term securities, less pension liabilities (in Germany), less intangibles. Adjusted book equity is book equity plus provisions plus deferred taxes less intangibles. In Panel C, Interest coverage ratio is income before interest and taxes divided by interest expense. Interest coverage is earnings before interest and taxes divided by interest.

Panel A: Extent of Leverage in Different Countries

Country	Number of Firms	Nonequity Liabilities to Total Assets (Medians (Means) Aggregate)		Debt to Total Assets (Medians (Means) Aggregate)		Debt to Net Assets (Medians (Means) Aggregate)		Debt to Capital (Medians (Means) Aggregate)		Interest Coverage Ratio (Medians (Means) Aggregate)	
		Book	Market	Book	Market	Book	Market	Book	Market	EBIT/Interest	EBITDA/Interest
United States	2580	0.58 (0.66)	0.44 (0.44)	0.27 (0.31)	0.30 (0.34)	0.34 (0.38)	0.24 (0.27)	0.37 (0.37)	0.28 (0.32)	2.41	4.05
Japan	69	0.69	0.49	0.37	0.26	0.44	0.29	0.53	0.34	2.19	3.66
Germany	514	0.69 (0.67)	0.45 (0.45)	0.35 (0.35)	0.22 (0.24)	0.48 (0.35)	0.27 (0.29)	0.53 (0.52)	0.29 (0.31)	2.46	4.66
France	191	0.75	0.55	0.42	0.31	0.58	0.39	0.61	0.41	2.39	3.71
Germany	191	0.73 (0.72)	0.60 (0.56)	0.16 (0.20)	0.12 (0.16)	0.21 (0.25)	0.15 (0.19)	0.38 (0.39)	0.23 (0.28)	3.20	6.81
France	225	0.71 (0.69)	0.64 (0.61)	0.25 (0.26)	0.21 (0.23)	0.39 (0.39)	0.32 (0.33)	0.48 (0.46)	0.41 (0.41)	2.64	4.35
Italy	118	0.78	0.68	0.29	0.23	0.43	0.34	0.57	0.45	2.15	3.47
United Kingdom	608	0.70 (0.67)	0.70 (0.67)	0.27 (0.28)	0.29 (0.28)	0.38 (0.38)	0.38 (0.38)	0.47 (0.46)	0.46 (0.47)	1.81	3.24
Canada	318	0.73	0.73	0.30	0.30	0.43	0.43	0.53	0.52	1.55	2.62
		0.56 (0.57)	0.40 (0.42)	0.18 (0.21)	0.14 (0.16)	0.26 (0.31)	0.18 (0.21)	0.28 (0.29)	0.19 (0.23)	4.79	5.44
		0.56	0.42	0.24	0.18	0.32	0.22	0.34	0.24	3.98	5.29
		0.26 (0.61)	0.49 (0.47)	0.32 (0.36)	0.28 (0.27)	0.37 (0.39)	0.32 (0.31)	0.39 (0.39)	0.35 (0.36)	1.55	3.05
		0.64	0.55	0.38	0.33	0.44	0.37	0.50	0.45	1.19	2.25

Table III—Continued

Panel B: Extent of Adjusted Leverage in Different Countries								
Country	Nonequity Liabilities to Total Assets (Medians (Aggregate))		Debt to Total Assets (Medians (Aggregate))		Debt to Net Assets (Medians (Aggregate))		Debt to Capital (Medians (Aggregate))	
	Book	Market	Book	Market	Book	Market	Book	Market
United States	0.52 (0.61)	0.42 (0.45)	0.25 (0.33)	0.16 (0.24)	0.32 (0.41)	0.20 (0.28)	0.33 (0.45)	0.23 (0.31)
Japan	0.62 (0.69)	0.37 (0.48)	0.21 (0.30)	0.12 (0.21)	0.33 (0.44)	0.16 (0.27)	0.37 (0.49)	0.17 (0.28)
Germany	0.69 (0.58)	0.41 (0.46)	0.11 (0.05)	0.10 (0.04)	0.17 (0.07)	0.13 (0.05)	0.18 (0.10)	0.15 (0.06)
France	0.69 (0.75)	0.59 (0.65)	0.18 (0.22)	0.13 (0.18)	0.32 (0.34)	0.25 (0.27)	0.34 (0.46)	0.28 (0.36)
Italy	0.68 (0.69)	0.68 (0.69)	0.21 (0.21)	0.21 (0.21)	0.33 (0.33)	0.31 (0.33)	0.39 (0.41)	0.36 (0.41)
United Kingdom	0.47 (0.48)	0.58 (0.35)	0.10 (0.13)	0.08 (0.09)	0.16 (0.38)	0.10 (0.11)	0.16 (0.12)	0.11 (0.13)
Canada	0.48 (0.56)	0.49 (0.50)	0.32 (0.36)	0.26 (0.32)	0.36 (0.41)	0.30 (0.36)	0.37 (0.44)	0.32 (0.40)

Table III, Panel A: Extent of leverage in different countries.

Table III, Panel B: Extent of adjusted leverage in different countries.

Table III—Continued

Panel C: Adjusted Leverage for Different Subsamples

Country	Debt level for Consolidated Firms in 1986 (Median)	Debt to Book Capital for Consolidated Firms of Similar Size Across Countries (Median)	The effect of consolidation on Debt to Book Assets (Median)	Comparisons with Studies Using OECD Data: Nonequity Liabilities to Book Assets (Aggregate)	Comparison with OECD Data: Debt to Book Assets in 1982 (Aggregate)	Changes in Leverage over Time for Consolidated Firms Reporting through 1982-91
		Debt to the size of the All	Debt to the size of the All			
		Debt to the size of the All	Debt to the size of the All			
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		Debt to the size of the All	Debt to the size of the All			
		Debt to the size of the All	Debt to the size of the All			

Table III, Panel C: Adjusted leverage for different subsamples.

Table IV
Sources of External Financing (1984-1991)

External financing as a fraction of total financing is the ratio of net external financing to the sum of cashflow from operations and net external financing. The figures for the various components of external financing are aggregated across all non-financial companies in the country and across all years, then normalized by the net external financing obtained by firms in the country during the period 1984-1991. Net debt financing is the sum of net short term debt issuances and long term debt issuances less long term debt reduction. Equity issuance includes the issue of both common and preferred stock and conversions of debt to equity. Net equity financing is the sum of equity issuance less equity reduction. The data are obtained from the flow of funds statement. The number of firms varies year by year.

Country	External Financing as a Fraction of Total Financing		Composition of External Financing			
			Net Debt Issuance		Net Equity Issuance	
	Global Vantage	OECD* Data	Global Vantage	OECD Data	Global Vantage	OECD Data
United States	0.20	0.23	1.02	1.34	-0.02	-0.34
Japan	0.50	0.56	0.80	0.85	0.20	0.15
Germany		0.33		0.87		0.13
France		0.35		0.39		0.61
Italy		0.33		0.65		0.35
United Kingdom	0.36	0.49	0.55	0.72	0.45	0.28
Canada	0.30	0.42	0.62	0.72	0.38	0.28

* OECD data from 1991-1993 editions of the OECD publication "Financial Statements of

Table IV: Sources of external financing, 1984-1991.

6.4 Table IV: Sources of external financing

Read:

- Table IV reports sources external financing as a fraction of total financing and decomposes external financing into net debt issuance and net equity issuance.
- U.S. firms rely relatively less on external financing than Japanese firms.
- Net debt issuance is generally the dominant component of external financing.
- Net equity issuance is small and can even be negative depending on the country and data source.

Economic message: Similar leverage levels can coexist with different financing flows. Capital structure is a stock concept, while external financing is a flow concept.

6.5 Table V: Tax treatment

Read:

- Table V calculates the tax advantage of debt relative to retained earnings and dividends under different assumptions about investors.
- The ranking of countries changes depending on whether the marginal investor is tax-free, high-tax, or average.
- The table shows that corporate taxes alone are insufficient for measuring debt tax incentives.
- Personal taxes and local taxes can materially change the effective tax advantage of debt.

Economic message: Tax-based capital-structure predictions are highly sensitive to the assumed marginal investor and tax base.

Table V
Tax Treatment of Interest, Dividends, and Retained Earnings
in Different Countries

Tax advantage of debt with respect to retained earnings (dividends) computed using the formulas derived by Miller (1977). The different rows correspond to different assumptions on the marginal personal tax rate of the investor. The "average" investor is assumed to head a family of three, and has three times the per capita income.

	United States	Japan	Germany	France	Italy	United Kingdom	Canada
Tax free investor							
Tax adv. w.r.t. retained earnings ^a	28.00	37.50	50.00	37.00	36.00	35.00	38.00
Tax adv. w.r.t. dividends ^a	28.00	37.50	36.00	42.00	36.00	35.00	38.00
Investor in the top tax bracket in 1990							
Tax adv. w.r.t. retained earnings	28.00	22.66	-6.38	-8.00	8.57	35.00	31.67
Tax adv. w.r.t. retained earn. no cap. gains tax ^c	8.33	21.88	-6.38	-26.00	8.57	-8.33	12.68
Tax adv. w.r.t. dividends	28.00	49.22	12.91	23.66	28.57	13.33	26.15
Investor in the top bracket in 1990 including local taxes							
Tax adv. w.r.t. retained earnings	40.25	40.15	14.89	-8.00	23.39	35.00	32.72
Tax adv. w.r.t. retained earn. no cap. gains tax	6.57	39.55	14.89	-28.57	23.39	-8.33	-3.55
Tax adv. w.r.t. dividends	40.25	60.71	30.32	23.66	40.15	13.33	35.32
Average Investor including local taxes							
Tax adv. w.r.t. retained earnings	40.25	40.15	37.50	27.70	23.39	35.00	20.79
Tax adv. w.r.t. retained earn. no cap. gains tax	7.79	39.55	37.50	13.93	23.39	-8.33	-3.55
Tax adv. w.r.t. dividends	40.25	59.01	30.34	13.00	28.17	13.33	35.32
Investor in the top bracket in 1983 including local taxes							
Tax adv. w.r.t. retained earnings	12.92	32.98	24.50	15.00	26.68	-8.39	27.14
Tax adv. w.r.t. retained earn. no cap. gains tax	-8.86	32.98	24.50	0.00	26.68	-54.84	2.37
Tax adv. w.r.t. dividends	51.69	44.37	34.27	47.51	63.34	31.43	27.85

Sources: various editions of "Doing Business in . . .," "Individual Taxes: A Worldwide Summary," and "Corporate Taxes: A Worldwide Summary" published by Price Waterhouse.

^a $[1 - (1 - t_p)(1 - t_{pE})/(1 - t_p)]$ where t_e is the corporate tax rate, t_{pE} the capital gain tax rate and t_p the personal tax rate.

^b $[1 - (1 - t_p)(1 - t_{pD})/(1 - t_p)]$ where t_{pD} is the tax rate on dividends after dividend tax credit is accounted for.

^c $[1 - (1 - t_p)/(1 - t_p)]$.

Table V: Tax treatment of interest, dividends, and retained earnings.

Table VI
The Allocation of the Pre-Tax Dollar to Various Routes and Changes in the Allocation Over Time

The aggregate interest expense, dividends, and earnings for an economy are computed by summing the individual firm values across firms. The share of a pre-tax dollar paid to debt in the economy is the interest paid (all variables are aggregates) divided by income before interest and taxes. The share of a pre-tax dollar paid through dividends is the dividends paid grossed up to a pre-tax value divided by income before interest and taxes. The dividends are grossed up to a pre-tax value by multiplying by earnings after interest and before taxes and dividing by earnings after interest and after taxes. The share of a pre-tax dollar retained is one minus the share paid to debt minus the share paid in dividends.

	United States		Japan	Germany	France	Italy	United Kingdom		Canada	
	1982-1984	1989-1991	1982-1984	1989-1991	1982-1984	1989-1991	1982-1984	1989-1991	1982-1984	1989-1991
Share of a pre-tax dollar paid through the route										
Debt	0.26	0.40	0.46	0.43	0.33	0.25	0.57	0.39	0.21	0.23
Dividends	0.39	0.16	0.16	0.37	0.32	0.25	0.21	0.18	0.27	0.26
Capital Gains	0.35	0.21	0.38	0.39	0.30	0.42	0.11	0.53	0.40	0.23
Route most tax advantaged by tax reforms between 1983 and 1990*										
Debt										
Dividends										
Retained earnings										
Route least tax advantaged by tax reforms between 1983 and 1990*										
Debt										
Dividends										
Retained earnings										
Change in share of pre-tax dollar flow between 1982-1984 and 1989-1991 allocated by companies consolidated and reporting throughout to route most tax advantaged										
Debt	0.14		-0.03		0.12		-0.11		0.09	
Dividends										
Retained earnings										
Change in share of pre-tax dollar flow between 1982-1984 and 1989-1991 allocated by companies consolidated and reporting throughout to route least tax advantaged										
Debt	-0.14		0.02		-0.07		-0.18		-0.06	
Dividends										
Retained earnings										

* From Table V under the assumption that capital gains tax is paid at the statutory rate.

Table VI: Allocation of the pre-tax dollar and changes over time.

6.6 Table VI: Allocation of the pre-tax dollar

Read:

- Table VI asks whether tax reforms are associated with changes in the route through which firms channel pre-tax income.
- The routes are debt interest, dividends, and retained earnings/capital gains.

- The evidence suggests that flows move somewhat toward the route that became more tax advantaged and away from the route that became less tax advantaged.
- The evidence is suggestive rather than definitive because it is based on a small number of countries and aggregate flows.

Economic message: Taxes appear to matter for financing flows, but the evidence is not strong enough to reduce capital structure to a simple tax story.

6.7 Table VII: Bankruptcy code across countries

Read:

- Table VII summarizes liquidation procedures, reorganization procedures, management control, automatic stays, and secured-creditor rights.
- The United States and Canada have relatively debtor-friendly reorganization channels.
- Germany is more creditor-friendly, with no stay for secured creditors.
- The United Kingdom gives secured creditors strong control through receivership and administrative receivership.
- These differences are economically meaningful, but they do not map cleanly into observed leverage rankings.

Economic message: Bankruptcy institutions shape the costs and control rights attached to debt, but the data do not support a simple monotonic relationship between creditor rights and leverage.

Country	Forms of Liquidation	Forms of Reorganization	Management Control in Bankruptcy	Automatic Stay	Rights of Secured Creditors
United States	Chapter 7: Can be voluntary (management files) or involuntary (creditors file).	Chapter 11: Can be voluntary (management files) or involuntary (creditors file).	Trustee appointed in Chapter 7. Management stays in control in Chapter 11.	Automatic stay on any attempts to collect debt once filing takes place.	Secured creditors get highest priority in any settlement. However, their attempts to collect payment are also stayed unless court or trustee approves.
Japan	Court Supervised Liquidation (<i>Shosen</i>) and Special Liquidation (<i>Tokubetsu Seisan</i>). The latter is less costly and a broader set of firms are eligible to file.	Composition (<i>Wagi-ho</i>), Corporate Arrangement (<i>Kaisha Seiri</i>) and Reorganization (<i>Kaisha Kosei-ho</i>). The list is in order of increasing eligibility. Only debtors file.	Third party is appointed except in composition and corporate arrangement.	All creditors are stayed except in court supervised liquidation and composition where only unsecured creditors are stayed.	Secured creditors have highest priority and greater voting rights in renegotiation. However, can be subject to stay depending on the petition that is filed.
Germany	Liquidation (<i>Konkursordnung</i>) can be requested by creditors or debtor. Management required to file as soon as it learns it is insolvent.	Composition (<i>Vergleich</i> or <i>Zwangvergleich</i>) can be filed for only by debtor.	Receiver appointed to manage firm.	Only unsecured creditors are stayed.	Secured creditors can recover their claims even after a bankruptcy filing. No stay for secured creditors.
France	Liquidation (<i>Liquidation Judiciaire</i>)	Negotiated Settlement (<i>Réglement Amiable</i>) where a court appointed conciliator attempts a settlement with creditors and Judicial Arrangement (<i>Redressement Judiciaire</i>).	Debtor loses control in liquidation. Debtor remains in control otherwise but submits to court appointed administrator's decisions in a judicial arrangement.	Stay on all creditors in judicial arrangement.	Secured creditors may lose status if court determines the security is necessary for continuation of the business, or if the securing asset is sold as part of settlement.

Italy	Bankruptcy (<i>Fallimento</i>).	Preventive Composition (<i>Concordato Preventivo</i>).	Debtor is removed from control over the firm.	Stay on all creditors.	Secured creditors stayed in bankruptcy, though composition allowed only if enough value exists to pay secured creditors in full and 40% of unsecured creditor claims. Secured creditors follow administrative claims in priority.
United Kingdom	Members' voluntary winding up, Creditors' voluntary winding up, Compulsory winding up.	Administration, Administrative Receivership (usually ends in sale of business), and Voluntary Arrangement.	Debtor is removed from control except in members' voluntary winding up.	Stay on all creditors in administration, on unsecured only in liquidation, and no stay in a voluntary arrangement until a proposal is approved.	Secured creditor may prevent administration order by appointing his own receiver. A creditor with a fixed or floating charge can appoint an administrative receiver to realize the security and pay the creditor.
Canada	Liquidation proceedings much like Chapter 7 in the United States.	Firms can file for automatic stay under the Companies Creditors Arrangement Act or the Bankruptcy and Insolvency Act.	Firm is in control in reorganizations while trustee is appointed for liquidations. Trustee may be appointed to oversee management in some reorganizations at the discretion of the court.	Stay on all creditors in reorganization.	Secured creditors have to give 10 days notice to debtor of intent to repossess collateral. Repossession even close to bankruptcy filing is permitted, but stayed after filing.

Sources: Kaiser (1994), Lo Pucki and Triantis (1994), Teichner (1991), and White (1993).

Table VII, part 1: Salient features of bankruptcy codes.

Table VII, part 2: Continued bankruptcy-code comparison.

6.8 Tables VIII and IX: Financial systems and leverage determinants

Read Table VIII:

- Table VIII reports domestic bank credit, stock market capitalization, and bond market capitalization in 1986.

- Bank credit is especially large in Japan and Germany.
- Stock markets are especially large in Japan, the United Kingdom, Canada, and the United States.
- The U.S. bond market is much larger than other G-7 corporate bond markets.

Read Table IX:

- Table IX estimates within-country leverage regressions.
- Tangibility is positively related to leverage in almost all countries.
- Market-to-book is generally negatively related to leverage.
- Size is usually positively related to leverage, except in Germany.
- Profitability is generally negatively related to leverage, except in Germany.

Economic message: Bank-versus-market orientation does not by itself explain leverage, but firm-level capital-structure correlations appear surprisingly stable across countries.

Table VIII
Size of Capital Markets in the G-7 Countries in 1986

Country	Domestic Bank Credit to the Private Sector as a Fraction of GDP (%)	Stock Market Capitalization (\$ billion)	Stock Market Capitalization as a Fraction of GDP (%)	Bond Market Capitalization (\$ billion)	Bond Market Capitalization as a Fraction of GDP (%)
United States	70.90	2128.00	49.85	993.20	23.27
Japan	104.22	1794.29	85.31	99.62	4.74
Germany	86.58	257.68	25.79	1.34	0.13
France	80.03	153.42	19.54	44.18	5.63
Italy	33.04	140.24	21.17	4.48	0.68
United Kingdom	53.85	472.90	83.70	14.01	2.48
Canada	44.21	185.20	50.56	27.17	7.42

Sources: Stock Market and GDP data are from Morgan Stanley Capital International Perspective, Domestic Credit from International Financial Statistics, and Bond Market Data are from Salomon Brothers International Bond Markets Analysis, 1992.

mies.²⁴ This distinction is also supported by other measures of the importance

Table VIII: Size of capital markets in the G-7 countries.

7 Short summary

- The paper studies capital structure in the G-7 countries using firm-level Global Vantage data.
- It asks whether U.S. capital-structure regularities generalize internationally.

Table IX
Factors Correlated with Debt to Book and Market Capital

The dependent variable is book leverage which is adjusted debt to adjusted debt plus book value of adjusted equity in 1991. Tangibility is the ratio of fixed assets to the book value of total assets. Market-to-book is the ratio of the book value of assets less the book value of equity plus the market value of equity all divided by the book value of assets. Logsales is the logarithm of net sales. Profitability is EBITDA divided by book value of assets. All the explanatory variables are four year averages (1987–90). Standard errors are in parentheses. The regression includes an intercept whose coefficient is not reported. The regression is estimated using maximum likelihood and a censored Tobit model. The estimated model is: $\text{Leverage}_{\text{Firm } i} = \alpha + \beta_1 \text{Tangibility}_i + \beta_2 \text{Market-to-book Ratio}_i + \beta_3 \text{Log Sales}_i + \beta_4 \text{Profitability}_i + \epsilon_i$.

Country Variable	United States	Japan	Germany	France	Italy	United Kingdom	Canada
Panel A: Book Capital							
Tangibility	0.50*** (0.04)	1.41*** (0.18)	0.42** (0.19)	0.53** (0.26)	0.36 (0.23)	0.41*** (0.07)	0.26*** (0.10)
Market-to-book	-0.17*** (0.01)	-0.04 (0.04)	-0.20*** (0.07)	-0.17** (0.08)	-0.19 (0.14)	-0.13*** (0.03)	-0.11*** (0.04)
Logsales	0.06*** (0.01)	0.11*** (0.02)	-0.07*** (0.02)	0.02 (0.02)	0.02 (0.03)	0.026*** (0.01)	0.08*** (0.01)
Profitability	-0.41*** (0.1)	-4.26*** (0.60)	0.15 (0.52)	-0.02 (0.72)	-0.16 (0.85)	-0.34 (0.30)	-0.46** (0.22)
Number of observations	2079	316	175	117	96	522	264
Pseudo R ²	0.21	0.29	0.12	0.12	0.05	0.18	0.19
Panel B: Market Capital							
Tangibility	0.33*** (0.03)	0.58*** (0.09)	0.28* (0.17)	0.18 (0.19)	0.48** (0.22)	0.27*** (0.06)	0.11 (0.07)
Market-to-book	-0.08*** (0.01)	-0.07*** (0.02)	-0.21*** (0.06)	-0.15** (0.06)	-0.18* (0.11)	-0.06** (0.03)	-0.13*** (0.03)
Logsales	0.03*** (0.00)	0.07*** (0.01)	-0.06*** (0.02)	-0.00 (0.02)	0.04 (0.03)	0.01 (0.01)	0.05*** (0.01)
Profitability	-0.6*** (0.07)	-2.25*** (0.32)	0.17 (0.47)	-0.22 (0.53)	-0.95 (0.77)	-0.47** (0.24)	-0.48*** (0.17)
Number of observations	2207	313	176	126	98	544	275
Pseudo R ²	0.19		0.14	0.28	0.12	0.19	0.30

*, **, and ***, significant at the 10, 5, and 1 percent level, respectively.

Table IX: Factors correlated with debt to book and market capital.

- A major part of the paper is devoted to making leverage measures comparable across accounting systems.
- After careful adjustments, leverage across G-7 countries is more similar than much prior work suggested.
- The United Kingdom and Germany are relatively less levered in many measures.
- Aggregate leverage does not split cleanly into bank-oriented versus market-oriented financial systems.
- External financing flows differ more than leverage stocks; Japanese firms rely more on external financing than U.S. firms.
- Tax incentives are hard to compare because they depend on both corporate and personal taxes and on the identity of the marginal investor.
- Bankruptcy procedures vary substantially across countries, but the relationship between bankruptcy law and leverage is not simple.
- Tangibility, market-to-book, size, and profitability are correlated with leverage in similar ways across many countries.
- The robustness of the correlations is impressive, but the underlying theory remains unsettled.

8 Conclusion

8.1 Core conclusion

Rajan and Zingales show that the main empirical determinants of leverage identified in U.S. data also appear in other G-7 countries. The evidence suggests that capital-structure choices are shaped by robust firm characteristics, but it does not validate a single dominant theory.

8.2 Economic intuition

Tangible assets support borrowing because they can serve as collateral. High market-to-book firms may avoid debt because growth opportunities are more vulnerable to debt overhang and financial distress. Large firms may borrow more because they are more diversified. Profitable firms may borrow less because they have internal funds. These stories are plausible, but the paper shows that each variable can have more than one interpretation.

8.3 Incremental contribution revisited

The paper's major contribution is not simply adding foreign countries to a U.S. regression. It shows how difficult capital-structure measurement is across accounting regimes and how easily institutional narratives can become too simple. The international setting provides a robustness test, but also reveals the limits of existing theories.

8.4 Implications

For empirical researchers, the paper implies that leverage should be measured carefully and that results should be checked under several definitions. For theorists, it implies that models must explain not just U.S. evidence, but the broad international stability of firm-level correlations. For policy discussions, it cautions against assuming that legal or tax reforms mechanically determine aggregate leverage.

8.5 Limitations

The analysis is observational and based on large listed firms. It cannot fully separate supply-side financial-system effects from firm demand for debt. Accounting adjustments reduce, but do not eliminate, measurement problems. The institutional analysis is rich but not a causal identification strategy.

8.6 Takeaway

The enduring takeaway is that we know some robust empirical facts about capital structure, but we know less about why those facts hold. Capital-structure determinants are similar across the G-7, yet the theory that maps those determinants to taxes, distress costs, institutions, and information problems remains incomplete.